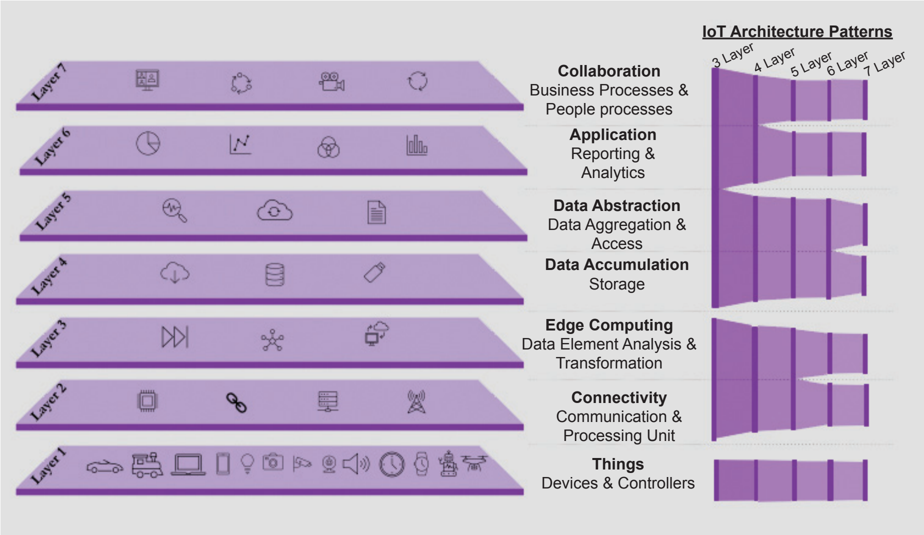

Figure 1: IoT architecture reference model and layered patterns evolution

services and device-oriented interactions are made possible by DSA. A network architect can divide functionality across separate computing resources using this architecture. The system can be scalable, fault-tolerant, and utilise all computing resources available to it from the edge, the data centre, the cloud, and everything in between thanks to a network topology

Advancements in enterprise 5G networks have increased the reliability of IoT architecture and these networks now process data faster than ever before. 5G addresses the implementation difficulties IoT architectures faced with spotty cellular networks, unreliable bandwidth, speed limits or unknown cost forecasting. Although designs for IoT architecture are maturing faster than connectivity speed, the improved speed has given use cases across healthcare, agriculture and manufacturing a renewed purpose.

The rapid adoption of IoT devices is just one dimension of how hungry consumers of the technology are for smarter devices, creating a significant uptick in the development and deployment of IoT device-related technologies. Let's hope that in the coming days there is a rapid unification of standards for leveraging the distributed services architecture of IoT, making its development and adoption smoother for everyone involved. END

The 6-layer architecture breaks down the network connectivity layer, creating an edge layer that enables efficient transport of data closer to the device. This improves the performance and utilisation of the device, and increases its speed and accuracy, specifically in transportation and logistics types of use cases.

Lastly, the most advanced IoT architecture model at present is the 7-layer architecture, which separates data accumulation and data abstraction. With this separation, the cloud plays a key role in the ever-increasing data collection journey of IoT devices, especially in the case of enterprises that deploy a massive number of devices for their customers and consumers. In data abstraction, layer data aggregation and access management for the application are primary concerns. In the data accumulation layer, the storage of data is the primary focus as it helps classify the quality and quantity of data stored by usage.

The evolution of IoT architecture along with commercial and open source solutions has made possible what was unimaginable just a few years ago.

Distributed services architecture

Distributed services architecture (DSA) enables smart communication, logic, and applications at all layers of the IoT infrastructure. The intent is to combine the various devices, services, and applications into a structured and flexible real-time data model. Open source DSA aims to create a network of manufacturers, creators, and solution providers who will add to a growing collection of distributed services that enable protocol translation and data integration to and from external data sources.

DSA is designed to connect upstream rather than downstream, making it easy to connect distributed services on a private and a public network. Decentralised

References

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